

brms / Stan, LOO, hierarchical models

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Write a simple multilevel model in **brms** syntax and describe each prior.
- Compare two candidate models using leave-one-out (LOO) cross-validation.
- Perform a prior predictive check and interpret it.

Prerequisites

Beta-binomial Bayesian intuition; linear mixed-effects models.

Background

brms is an interface that translates R formulas into Stan programs and runs Hamiltonian Monte Carlo to draw from the posterior. It is particularly strong for multilevel (hierarchical) models, where the benefits of Bayesian computation — full posterior uncertainty, proper propagation into predictions — show up most clearly.

LOO cross-validation, implemented via the **loo** package, estimates out-of-sample predictive accuracy from the posterior without refitting, using Pareto-smoothed importance sampling. It is a modern replacement for AIC/BIC in Bayesian model comparison, and comes with diagnostics that warn when the approximation is unreliable.

Compilation of Stan models is slow, and downloading the toolchain exceeds what we can assume in a rendering environment. We therefore show the **brms** code with `#! eval: false` and walk through prior predictive simulation and a small frequentist counterpart that runs end-to-end.

Setup

```
library(tidyverse)
library(lme4)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Fit a two-level hierarchical model to simulated clinical data — patient outcomes nested within centre — and compare a random-intercept against a random-slope model.

2. Approach

Simulate 20 centres with varying mean outcomes and varying dose effects.

```
centre <- rep(seq_len(n_centre), each = per_centre)
alpha0 <- rnorm(n_centre, 0, 1)
beta0 <- rnorm(n_centre, 0.5, 0.3)
dose <- rnorm(n_centre * per_centre)
```

```

y <- alpha0[centre] + beta0[centre] * dose + rnorm(n_centre * per_centre, 0, 0.7)
d <- tibble(y = y, dose = dose, centre = factor(centre))
ggplot(d, aes(dose, y, colour = centre)) +
  geom_point(alpha = 0.5) + geom_smooth(method = "lm", se = FALSE) +
  theme(legend.position = "none")

```

3. Execution

A classical `lmer` random-intercept-and-slope fit as a stand-in we can run.

```

fit_lmer_rs <- lmer(y ~ dose + (dose | centre), data = d)
summary(fit_lmer_rs)$varcor
anova(fit_lmer_ri, fit_lmer_rs)

```

Prior predictive simulation (frequentist-style) for the Bayesian counterpart we are about to sketch.

```

a <- rnorm(1, 0, 1); b <- rnorm(1, 0, 1); sd_a <- abs(rnorm(1, 0, 1))
a + b * dose + rnorm(length(dose), 0, sd_a)
})
ggplot(tibble(y = as.numeric(prior_sim)), aes(y)) +
  geom_histogram(bins = 40, fill = "grey60", colour = "white") +
  labs(x = "y drawn under priors only", y = "count")

```

The `brms` counterpart.

```

library(brms)
priors <- c(
  prior(normal(0, 2), class = "Intercept"),
  prior(normal(0, 1), class = "b"),
  prior(exponential(1), class = "sd"),
  prior(exponential(1), class = "sigma")
)
m_ri <- brm(y ~ dose + (1 | centre), data = d,
  prior = priors, chains = 2, iter = 2000, refresh = 0)
m_rs <- brm(y ~ dose + (dose | centre), data = d,
  prior = priors, chains = 2, iter = 2000, refresh = 0)
loo_compare(loo(m_ri), loo(m_rs))

```

4. Check

Compare to `lmer` and report pooled vs unpooled estimates.

```
fixef_lmer
```

5. Report

On simulated hierarchical data (20 centres $\times$ 15 patients), a random-slope `lmer` model recovered an average dose effect of `round(fixef_lmer["dose"], 2)` (SE from `lmer`), with substantial centre-to-centre variation. The `brms` sketch shown in the accompanying code would fit the same structure with explicit priors, and `loo_compare` would rank the random-slope model against the random-intercept model.

In practice, the Bayesian and frequentist fits should agree closely when priors are weakly informative and n is large; the Bayesian route adds explicit prior sensitivity and full posterior uncertainty for downstream prediction.

Common pitfalls

- Fitting a flat prior in a small-sample hierarchical model and getting a divergent posterior on variance components.
- Comparing LOO elpd differences smaller than about 4 SE without noting the uncertainty.
- Forgetting to scale continuous predictors; default priors assume normalised inputs.

Further reading

- Bürkner P-C (2017), *brms: An R package for Bayesian multilevel models using Stan*.
- Vehtari A, Gelman A, Gabry J (2017), *Practical Bayesian model evaluation using leave-one-out cross-validation*.

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)